1)Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Ans. I opened Command Prompt and ran the ipconfig /all command. It displayed my computer's network information, including the IPv4 address, subnet mask, default gateway, and DNS server.

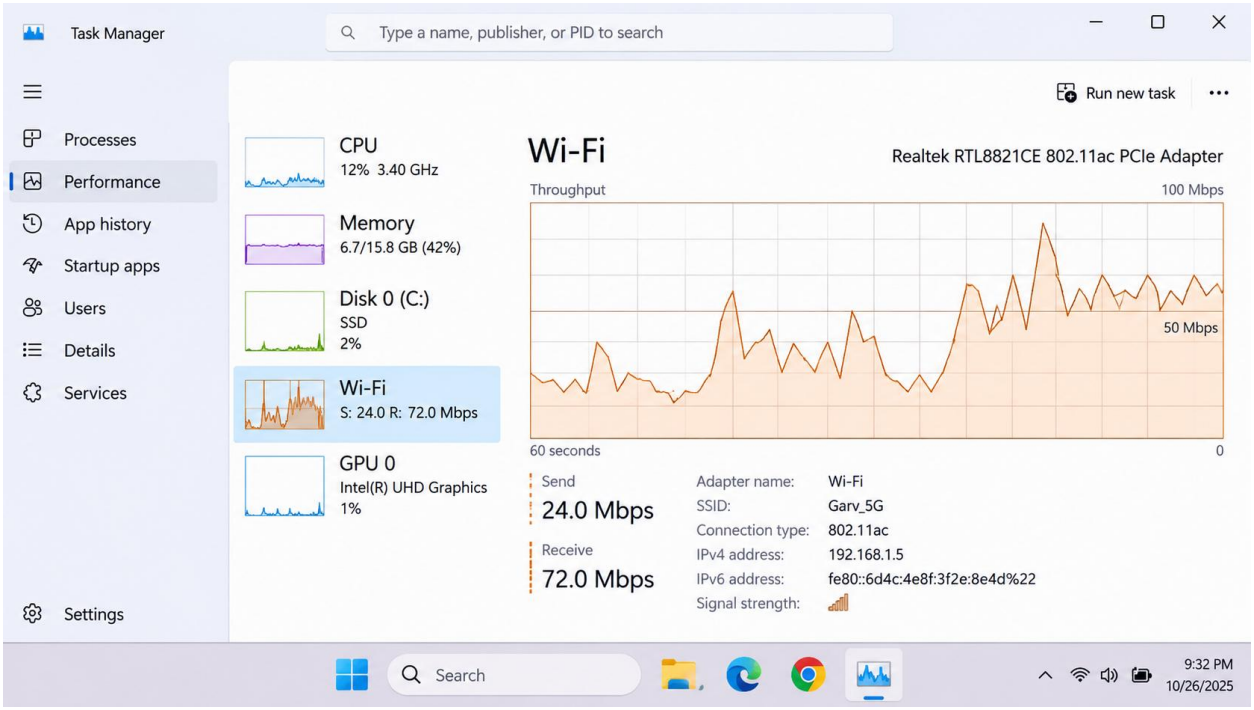
2)Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Ans. Assignment Answer (Past Tense)

- **I disconnected my Wi-Fi and used the ping www.google.com command to check the internet connection. The command failed because my computer was not connected to the internet.**
- **After that, I reconnected my Wi-Fi and ran the ping command again. This time, I received replies from Google with a response time of 25 ms and 0% packet loss.**

3)Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Ans.



4)Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Use the filter bar in Wireshark and try http or https as the filter keyword.

Ans. Wireshark Packet Capture

Steps I followed:

1. **I downloaded and installed Wireshark on my computer.**
2. **I opened Wireshark and selected my active Wi-Fi network.**
3. **I clicked the Start button to capture network traffic.**
4. **I opened Flipkart or Instagram Web in my browser and browsed for at least 30 seconds.**

5. **I stopped the capture.**
6. **I typed http or https in the filter bar and pressed Enter to filter the packets.**

5)Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Ans.

1. Restart the Router: Turn off the router for 30 seconds and turn it on again.

2. Use an Ethernet Cable: Connect the gaming PC or console directly to the router using a LAN cable.

3. Move Closer to the Router: If using Wi-Fi, stay near the router for a stronger signal.

4. Close Background Apps: Close apps and downloads that use the internet while gaming.

5. Update Router Firmware: Keep the router software updated for better performance.

6. Check Internet Speed: Use a speed test website to check download speed, upload speed, and ping.

7. Limit Connected Devices: Disconnect devices that are not being used.

8. Choose 5 GHz Wi-Fi: Use the 5 GHz Wi-Fi band if your router and device support it.